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import os

# Database file configuration
DATA_FILE = "student_records.txt"

def initialize_system():
    """Initializes the data file if it does not exist."""
    if not os.path.exists(DATA_FILE):
        with open(DATA_FILE, "w") as f:
            pass

def main_menu():
    while True:
        print("\n" + "="*45)
        print(" STUDENT RECORD MANAGEMENT SYSTEM (SRMS) ")
        print("="*45)
        print("1. Add New Student Record")
        print("2. Search Record by Roll Number")
        print("3. Update Existing Record")
        print("4. Display All Records")
        print("5. Exit System")
        print("="*45)
        choice = input("Select an option (1-5): ")

        if choice == '1':
            add_student()
        elif choice == '2':
            search_student()
        elif choice == '3':
            update_student()
        elif choice == '4':
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        display_all()
    elif choice == '5':
        print("\nSystem shutting down. Securely exiting...")
        break
    else:
        print("\n[ERROR] Invalid selection. Please choose between 1-5.")

def add_student():
    print("\n--- Registering New Student ---")
    try:
        roll = int(input("Enter Roll Number: "))
        name = input("Enter Full Name: ").strip()
        grade = input("Enter Grade (e.g., A, B+): ").strip().upper()
        info = input("Enter Personal Info (City/Contact): ").strip()

        # Check for duplicate roll number
        if os.path.exists(DATA_FILE):
            with open(DATA_FILE, "r") as f:
                for line in f:
                    if line.strip() and line.strip().split("|")[0] == str(roll):
                        print(f"\n[ERROR] Roll Number {roll} already exists.")
                        return

        # Using pipe (|) as a delimiter
        record = f"{roll}|{name}|{grade}|{info}\n"
        with open(DATA_FILE, "a") as f:
            f.write(record)
        print("\n[SUCCESS] Student record saved successfully.")

    except ValueError:
        print("\n[ERROR] Invalid input. Roll Number must be an integer.")
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def display_all():
    if not os.path.exists(DATA_FILE) or os.stat(DATA_FILE).st_size == 0:
        print("\n[INFO] No records found in the database.")
        return

    print("\n" + "-"*65)
    print(f"{'ROLL':<10} | {'NAME':<20} | {'GRADE':<10} | {'REMARKS'}")
    print("-"*65)

    with open(DATA_FILE, "r") as f:
        for line in f:
            if line.strip():
                parts = line.strip().split("|")
                if len(parts) == 4:
                    r, n, g, i = parts
                    print(f"{r:<10} | {n:<20} | {g:<10} | {i}")
                else:
                    print(f"[CORRUPTED RECORD] {line.strip()}")

    print("-"*65)

def search_student():
    search_id = input("\nEnter Roll Number to search: ").strip()
    found = False

    if os.path.exists(DATA_FILE):
        with open(DATA_FILE, "r") as f:
            for line in f:
                data = line.strip().split("|")
                if len(data) == 4 and data[0] == search_id:
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            print("\n[RECORD FOUND]")
            print(f"  Roll No : {data[0]}")
            print(f"  Name    : {data[1]}")
            print(f"  Grade   : {data[2]}")
            print(f"  Info    : {data[3]}")
            found = True
            break

    if not found:
        print(f"\n[NOT FOUND] No record exists for Roll Number: {search_id}")

def update_student():
    target_id = input("\nEnter Roll Number to update: ").strip()
    temp_records = []
    is_updated = False

    if not os.path.exists(DATA_FILE):
        print("\n[ERROR] Database file not found.")
        return

    with open(DATA_FILE, "r") as f:
        for line in f:
            data = line.strip().split("|")
            if len(data) == 4 and data[0] == target_id:
                print(f"\nCurrent Record:")
                print(f"  Name : {data[1]} | Grade : {data[2]} | Info : {data[3]}")
                data[1] = input("Enter New Name  (leave blank to keep): ").strip() or data[1]
                data[2] = input("Enter New Grade (leave blank to keep): ").strip().upper() or data[2]
                data[3] = input("Enter New Info  (leave blank to keep): ").strip() or data[3]
                is_updated = True
            temp_records.append("|".join(data) + "\n")
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    if is_updated:
        with open(DATA_FILE, "w") as f:
            f.writelines(temp_records)
        print("\n[SUCCESS] Record updated successfully.")
    else:
        print(f"\n[ERROR] Roll Number {target_id} not found in database.")

def delete_student():
    target_id = input("\nEnter Roll Number to delete: ").strip()
    temp_records = []
    is_deleted = False

    if not os.path.exists(DATA_FILE):
        print("\n[ERROR] Database file not found.")
        return

    with open(DATA_FILE, "r") as f:
        for line in f:
            data = line.strip().split("|")
            if len(data) == 4 and data[0] == target_id:
                confirm = input(f"Delete record for '{data[1]}'? (yes/no): ").strip().lower()
                if confirm == "yes":
                    is_deleted = True
                    continue  # Skip writing this record back
            temp_records.append(line)

    if is_deleted:
        with open(DATA_FILE, "w") as f:
            f.writelines(temp_records)
        print("\n[SUCCESS] Record deleted successfully.")
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    else:
        print(f"\n[ERROR] Roll Number {target_id} not found, or deletion cancelled.")


if __name__ == "__main__":
    initialize_system()
    main_menu()
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